Unit-2 Research Topic, Literature Survey and Problem Statement

Basanta Chapagain, Patan Multiple Campus

Research Theme

- A **research theme** is the broad area or overarching direction of a research study. It represents a major subject or field of knowledge within which multiple specific research topics can be explored and developed.
- A research theme usually reflects current trends, emerging interests, or significant challenges in a particular discipline. It serves as a general framework that allows researchers to investigate various issues connected to the same core idea.

Examples of Research Themes : Digital Transformation in Education , Human–Computer Interaction, Artificial Intelligence and Machine Learning, Cybersecurity and Data Privacy, Sustainable Information Technology

Research Theme

Characteristics of a Research Theme

- Very broad and general in scope
- Encompasses many related topics and sub-areas
- Commonly used in academic conferences, journals, and research funding programs
- Helps organize and group similar studies under one unified concept

Research Theme

Example in the Field of Computer Applications

- **Theme:** Cybersecurity

Possible Research Topics under this theme:

- Network security in cloud computing environments
- Development of phishing attack detection systems using machine learning
- Security vulnerabilities and protection mechanisms in mobile banking applications

Research Topic

- A **research topic** is the **specific subject or area that a researcher wants to study**. It is narrower than the theme and identifies **what exactly the researcher wants to investigate**.
- The topic provides the **direction and scope** of the research work.
- Choosing a research topic is often the most underrated hurdle in the academic process. It's the foundation of your entire project

"If the foundation is weak, the entire structure cannot remain stable—no matter how well the architecture or methodology is designed."

Prepared By: Basanta Chapagain
[academic.basanta@gmail.com]

Research Topic

Example Topics (Computer Applications)

- Impact of cloud computing on business data storage
- Use of mobile applications for online learning
- Performance analysis of web-based database systems
- Role of artificial intelligence in healthcare systems

Research Topic

Characteristics of a Good Research Topic

<u>A research topic should be:</u>

Clear and Focused

- In tech, "broad" topics lead to shallow results. You must define the specific technology, the user base, and the intended outcome.
- **The "Too Broad" Version:** "Artificial Intelligence in Healthcare."
- **The "Good" Topic:** "Comparing the accuracy of **Convolutional Neural Networks (CNN)** vs. **Vision Transformers** in detecting early-stage melanoma from smartphone images."
- **Why it works:** It identifies the specific algorithms, the exact medical condition, and the hardware constraints.

Research Topic

Interesting to the Researcher

- The field of Computer Applications moves fast. You should choose a sub-sector (e.g., Cybersecurity, UX/UI, Cloud Computing) that aligns with your career goals.
- **The Motivation:** If you are passionate about gaming, researching **"Latency reduction algorithms in Cloud Gaming"** will be much more rewarding than researching database management systems.
- **The Persistence Factor:** When your code refuses to compile or your model won't converge, your interest in the subject is what keeps you debugging.

Research Topic

Relevant to the Field of Study

- A good topic addresses "The State of the Art." It should tackle current bottlenecks or emerging paradigms in computing.
- **The "Knowledge Gap":** Many apps exist for productivity, but perhaps there is a gap in **"Cross-platform synchronization for AR-based collaborative workspaces."**
- **Industry Alignment:** Researching **"Edge Computing for IoT in Smart Agriculture"** is highly relevant as industry shifts away from centralized cloud processing to save bandwidth.

Research Topic

Feasible (Resources and Time)

- Feasibility in computing often comes down to hardware, data, and API access.
- **Data Availability:** You cannot research "Predicting stock market crashes with AI" if you don't have access to high-frequency trading datasets or the GPU power to process them.
- **Technical Scope:** Developing a full-scale decentralized social media platform might be too much for a 3-month project. However, developing a **"Proof-of-concept Smart Contract for secure user authentication on the Ethereum blockchain"** is feasible.

Research Topic

Supported by Existing Literature

- Even if you are using a brand-new framework (like the latest version of Flutter or a new LLM), your research must be grounded in established computer science principles.
- **Theoretical Grounding:** Your work on a new mobile app should be supported by existing literature on **Human-Computer Interaction (HCI)** or **Agile Development Methodologies**.
- **Benchmarking:** To prove your application is "good," you need existing studies to provide benchmarks (e.g., "The industry standard for app load time is $2/ seconds; your proposed optimization reduced it to $1.2/ seconds").

Research Issue / Research Problem

- A **research issue or research problem** refers to the **specific difficulty, gap, or unanswered question** within the research topic that needs investigation.
- It identifies **what problem exists and why the research is necessary**.
- The **research problem is the core focus of the study** because the research is conducted to **find solutions or explanations for this problem**.

A research problem usually arises when:

- Existing studies show **limitations**
- There is **conflicting evidence**
- A practical problem requires **scientific investigation**

Research Issue / Research Problem

Characteristics of a Good Research Problem

- Clearly defined
- Researchable and measurable
- Relevant to the field
- Significant enough to contribute knowledge
- Feasible to investigate

Research Issue / Research Problem :Example

Topic: **Online learning systems in universities**

Possible research problems:

- Low student engagement in virtual classrooms
- Difficulty in assessing student performance online
- Lack of interaction between teachers and students in e-learning

- Topic: **Mobile Banking Applications**

Possible research problems:

- Security risks in mobile banking systems
- Lack of user trust in digital payment systems

Research Issue / Research Problem :Example

Example in Computer Applications

- Theme: **Human–Computer Interaction**
- Topic: **User interface design for mobile applications**
- Research Problem: **How poor interface design affects user experience in mobile learning apps**

Research Issue / Research Problem :Example

Importance of Identifying Research Topic/Issue/Theme

- Identifying the research theme, topic, and issue is important because it:
- Provides **clear direction for the study**
- Helps define the **scope and objectives**
- Guides the **selection of research methods**
- Helps in **reviewing relevant literature**
- Ensures the research **addresses real problems**
- Without clearly identifying these elements, research may become **unfocused and ineffective**.

Research Issue / Research Problem :Steps

Step 1: Choose a Broad Area of Interest

- First, select a **general field** that interests you or is related to your academic discipline.

Examples in Computer Applications

- Artificial Intelligence
- Web Development
- Cybersecurity
- E-learning Systems
- Mobile Application Development

Example

- Broad Area: **E-Learning Systems**

Research Issue / Research Problem :Steps

Step 2: Conduct Preliminary Reading (Literature Review)

- Read **journals, conference papers, articles, and books** to understand:
- What research already exists
- Current trends and technologies
- Problems researchers have identified
- This helps you **avoid repeating existing research** and find areas that need further study.

Example

- After reading articles on e-learning systems, you may find that:
- Many universities use **Learning Management Systems (LMS)**.
- Students sometimes show **low engagement in online classes**.

Research Issue / Research Problem :Steps

Step 3: Identify Gaps or Issues

- A **research gap** is something that existing research has not fully addressed.
- Look for:
- Limitations in previous studies
- Unanswered questions
- New technologies not yet studied
- Problems occurring in real practice

Example

- Gap identified:
- Many studies discuss e-learning adoption, but **few examine student engagement in programming courses delivered online**.

Research Issue / Research Problem :Steps

Step 4: Narrow Down the Topic

- Convert the general area into a **more focused topic**.

Example

- Broad area: E-learning systems
  Focused topic: **Online learning platforms for programming courses**

Research Issue / Research Problem :Steps

Step 5: Define the Specific Research Problem

- Now convert the topic into a **clear research problem or issue**.

<u>Ask questions such as</u>:

- What exactly is the problem?
- Why does it occur?
- Who is affected?

Example Research Problem

- Students learning programming through online platforms often show **low engagement and difficulty understanding practical coding concepts**.

Research Issue / Research Problem :Steps

Step 6: Formulate Research Questions

- Transform the research problem into **specific research questions**.

Example Research Questions

- How does the design of an online learning platform affect student engagement in programming courses?
- What features improve student participation in online coding classes?

Research Issue / Research Problem :Steps

Step 7: Check Feasibility

Ensure the problem is **researchable** with available:

- Time
- Data
- Resources
- Technical tools

Example

- You can collect data by:
- Surveying computer science students
- Analyzing usage data from an LMS
- Conducting interviews with instructors

Research Issue / Research Problem :Tasks

- Select Theme
- Select Topic
- Write Research Problem and Research Questions?

Research Issue / Research Problem :Criteria

- **Researcher's Interest:** The research problem should be interesting to the researcher so that they remain motivated to study it.
- **Relevance to the Field:** The problem should be related to the researcher's academic or professional field of study.
- **Availability of Data:** Adequate data and information should be available to investigate the research problem.
- **Feasibility:** The research problem should be manageable within the available time, resources, and skills.
- **Researchability:** The problem must be capable of being studied using scientific research methods.

Research Issue / Research Problem :Criteria

- **Significance:** The research problem should contribute useful knowledge or help solve a real-world issue.
- **Clarity and Specificity:** The research problem should be clearly defined and not too broad or vague.
- **Originality:** The research problem should provide new insights or add value to existing knowledge.
- **Ethical Acceptability:** The research should follow ethical principles and not harm participants.
- **Social or Practical Value:** The research problem should have benefits for society, organizations, or the academic community.